

Project Overview

Task List:

After organizing the project work, we believe it can basically be divided into four tasks in total. Please choose one according to your own situation and personal interest. If too many people select the same one, we will discuss and adjust. Overall, I think the workload for each person will not differ much – the difference is mainly in focus. If you really have no preference, you can also be assigned randomly.

Data Cleaning (2 people needed)

Position in the project:

Responsible for defining the data standard of the temporal network, deciding the input content and how to process the input.

Task objectives:

Find specific datasets, design the data structure used in the project, and clarify the data types required for the user upload function.

Required learning:

Python; Python libraries: Streamlit, Pandas; JSON or CSV.

Integration with other groups:

- Provide a cleaned adjacency list or graph object for the BFS algorithm group.
 - Provide the initial data state in `st.session_state` for the web implementation group.
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BFS Algorithm (2 people needed)

Position in the project:

Handle logical computation and generate path instructions.

Task objectives:

Implement a time-aware BFS algorithm and define the node traversal logic displayed on the webpage.

Required learning:

Python; Python libraries: NetworkX; priority queue.

Integration with other groups:

- Obtain the graph data model from the data group.

- Pass algorithm results to the visualization group for dynamic rendering.
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Web Implementation (2 people needed)

Position in the project:

The interactive framework of the project, responsible for building the front-end interface.

Task objectives:

Build containers based on the UI design, manage current data, and implement required functions according to requirement analysis.

Required learning:

Python; Streamlit library.

Integration with other groups:

- Call functions from the algorithm group to trigger computation.
 - Receive layout suggestions from the UI design group.
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UI Design & Requirement Analysis (1 person needed)

Position in the project:

Page planning.

Task objectives:

Design the overall webpage and translate the supervisor's requirements into concrete functions.

Required learning:

Python; Streamlit library; visual expression.

Integration with other groups:

- Clarify the user interaction fields in the UI interface.
 - Determine component layout and functional requirements.
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Timeline

The situation before the 23rd is not considered for now. After receiving the teacher's feedback, the schedule may change at any time. This is only for reference.

From **February 23 to April 1** will be the development period. Teams will work according to the group assignments. Every Wednesday during this period will serve as a task milestone deadline, and currently we assume a meeting at **5 PM each Wednesday evening** (though this may change depending on progress and availability).

February 23 - March 11

Each group is expected to produce a preliminary report. During this time, everyone should first learn the tools and library methods we will use and determine the necessary specifications within each group.

You may need to produce a document so coding can proceed smoothly later – for example, what data format the data group will use, what fields the cleaned tables should contain, etc. Each group has the right to decide but also the responsibility to investigate and choose.

If anyone finds useful tools, feel free to share them in the group chat.

March 11 - April 1

This is the implementation phase. Using the agreed specifications, each group should complete the corresponding functions. There are about two weeks in total, and groups that need coordination should communicate directly with each other.

After April 1

We will prepare the demonstration video. Afterwards there may be many tasks such as fixing bugs, revisions, PPT preparation, and presentation scripts. The work mode may return to weekly task assignments.

Meeting Record Order

During the development period meetings (Feb 23 – Apr 1), records will be taken in alphabetical order:

Ce Wang
Li Bowen
Liu Yiqing
Ranjan Ankit
Tang Zixuan
Xu Xiaohan
Zeng Wenbin